⌈1-form.

⌈Def. A map $\varphi : \bigcup_{p\in\mathbb{R}^3} T_p\mathbb{R}^3 \to \mathbb{R}$ is called a 1-form if: for fixed $p\in\mathbb{R}^3$,

$$\varphi(a\cdot V_p + W_p) = a\cdot\varphi(V_p) + \varphi(W_p) \text{ for all } a\in\mathbb{R} \text{ and } V_p, W_p \in T_p\mathbb{R}^3.$$

That is, the restriction $\varphi_p : T_p\mathbb{R}^3 \to \mathbb{R} : V_p \mapsto \varphi(V_p)$ is a linear functional.⌋

⌈Def. Let $\varphi, \varphi_1, \varphi_2$ be 1-forms, and $f, g : \mathbb{R}^3 \to \mathbb{R}$ be functions, and $V, W : \mathbb{R}^3 \to \bigcup_{p\in\mathbb{R}^3} T_p\mathbb{R}^3$ be vector fields. Define the operations:

$$(\varphi_1 + \varphi_2)(V_p) \stackrel{\text{def}}{=} \varphi_1(V_p) + \varphi_2(V_p) \quad \text{and} \quad (f\varphi)(V_p) \stackrel{\text{def}}{=} f(p)\cdot\varphi(V_p)$$

$$\varphi(fV + gW) \stackrel{\text{def}}{=} f\varphi(V) + g\varphi(W) \quad \text{and} \quad (f\varphi_1 + g\varphi_2)(V) = f\varphi_1(V) + g\varphi_2(V)$$

⌈Def. Given a differentiable function $f : \mathbb{R}^3 \to \mathbb{R}$, the 1-form

$$df : \bigcup_{p\in\mathbb{R}^3} T_p\mathbb{R}^3 \to \mathbb{R}^3 : V_p \mapsto V_p[f]$$

is called the differential of f.

Further, for the projections $x_i : \mathbb{R}^3 \to \mathbb{R} : (x_1, x_2, x_3) \mapsto x_i$ $(i = 1, 2, 3)$, the set $\{(dx_1)_p, (dx_2)_p, (dx_3)_p\}$ is a dual basis for $T_p\mathbb{R}^3$ corresponding to $\{(e_1)_p, (e_2)_p, (e_3)_p\}$ for each $p\in\mathbb{R}^3$.

⌈Lemma. Given 1-form φ on $\mathbb{R}^3$,

$$\varphi = \sum_{i=1}^{3} \varphi\circ e_i \cdot dx_i.$$

In particular, $df = \sum_{i=1}^{3} df\circ e_i \cdot dx_i = \sum_{i=1}^{3} \frac{\partial f}{\partial x_i}\cdot dx_i$

⌈Example.

Let $f(x, y, z) = (x^2 - 1)y + (y^2 + 2)z$. Then,

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$$

$$= 2xy\,dx + (x^2 + 2yz - 1)\,dy + (y^2 + 2)\,dz.$$

So, $V_p[f] = df(V_p) = 2p_1p_2v_1 + (p_1^2 + 2p_2p_3 - 1)v_2 + (p_2^2 + 2)v_3$.

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So $\quad V_p[f] = df(V_p) = 2p_1p_2v_1 + (p_1^2 + 2p_2p_3 - 1)v_2 + (p_2^2+2)v_3.$